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SERVER, CONTROL METHOD, STORAGE MEDIUM, AND PRINTING SYSTEM

Abstract

A server providing a cloud print service includes at least one processor that executes a set of instructions, when executed, causing the server to perform operations including acquiring attribute information indicating printing capabilities of a printing apparatus when the printing apparatus is registered as an output destination printer, registering output destination printer information by associating the acquired attribute information with information indicating an output destination, and transmitting, in a case where a request for acquiring capability information about the registered output destination printer is received, the attribute information about the registered output destination printer to a client terminal as a response to the request, wherein in the registering, even in a case where the acquired attribute information about the printing apparatus includes an attribute that is not supported by the cloud print service, the attribute information is registered as the output destination printer information without discarding the attribute information.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation of U.S. patent application Ser. No. 18/509,155, filed Nov. 14, 2023, which is a Continuation of U.S. patent application Ser. No. 17/820,331, filed Aug. 17, 2022, now U.S. Pat. No. 11,847,373, which is a Continuation of U.S. patent application Ser. No. 17/227,076, filed Apr. 9, 2021, now U.S. Pat. No. 11,461,064, which claims the benefit of Japanese Patent Application No. 2020-073487, filed Apr. 16, 2020, all of which are hereby incorporated by reference herein in their entirety.

BACKGROUND
Field of the Disclosure
[0002] The present disclosure relates to a server, a control method, a storage medium, and a printing system for providing a print service.
Description of the Related Art
[0003] In cloud print services that have been widely used in recent years, a print job is input and then transmitted to a printing apparatus via the cloud, as discussed in Japanese Patent Application Laid-Open No. 2012-133489. In such a printing system, an administrator first registers a printing apparatus in the cloud print service to which the administrator belongs. In addition, cloud print services provided by information technology (IT) vendors and the like are desired to be compatible with various types of printing apparatuses in view of user-friendliness. In such cloud print services, capability information defined in a general-purpose printing technique, such as an Internet Printing Protocol (IPP), is acquired from a registered printing apparatus, and the capabilities of the printing apparatus are managed.
[0004] Subsequently, the administrator sets a user to be permitted to use the printing apparatus. The user permitted to use the printing apparatus accesses the cloud print service from a client terminal, such as a personal computer (PC), and selects a printing apparatus to be used for printing from among the printing apparatuses registered on the cloud print service. Then, the cloud print service notifies the client terminal of the capability information about the printing apparatus. In this case, capability information that is not supported by a print client of the client terminal is discarded.
[0005] The client terminal manages the capability information about the output destination printer based on the notified capability information. Upon receiving a user operation for changing print settings, the client terminal generates and displays a print setting screen based on the managed capability information. This enables the user to make print settings on the client terminal depending on the capabilities of the printing apparatus. Then, when a printing instruction is made by a user of an information processing apparatus, the information processing apparatus transmits print data to the cloud print service. The cloud print service stores the received print job in a storage. The

printing apparatus acquires the print job stored in the cloud print service, and executes printing. Printing is executed via the cloud in a series of processing described above.
[0006] Examples of cloud print services include Google Cloud Print®, Microsoft Hybrid Cloud Print®, and uniFLOW Online®.
SUMMARY
[0007] According to embodiments of the present disclosure, a server providing a cloud print service includes at least one memory that stores a set of instructions and at least one processor that executes the instructions, when executed, causing the server to perform operations including acquiring attribute information indicating printing capabilities of a printing apparatus when the printing apparatus is registered as an output destination printer for the cloud print service, registering output destination printer information by associating the acquired attribute information with information indicating an output destination, and transmitting, in a case where a request for acquiring capability information about the registered output destination printer from a client terminal is received, the attribute information about the registered output destination printer to the client terminal as a response to the request, wherein in the registering, even in a case where the acquired attribute information about the printing apparatus includes an attribute that is not supported by the cloud print service, the attribute information is registered as the output destination printer information without discarding the attribute information.
[0008] Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0009] FIG. **1** is a diagram illustrating an example of a printing system.
[0010] FIG. **2** is a sequence diagram illustrating an operation of a cloud print service (CPS) of a related art.
[0011] FIG. **3** illustrates an example of a print setting screen for the cloud print service of the related art.
[0012] FIG. **4** is a table illustrating differences in supported print capabilities.
[0013] FIG. **5** is a block diagram illustrating an example of a hardware configuration of a printing apparatus.
[0014] FIG. **6** is a block diagram illustrating an example of a hardware configuration of each of a server and client terminals.
[0015] FIG. **7** is a sequence diagram illustrating an example of each of printer registration processing and print processing.
[0016] FIG. **8** illustrates an example of a print setting screen to be displayed on an operation unit of a client terminal.
[0017] FIG. **9** is a flowchart illustrating an example of registration control processing to be executed by the printing apparatus.
[0018] FIG. **10** is a flowchart illustrating an example of registration control processing to be executed by a CPS.
[0019] FIG. **11** is a flowchart illustrating an example of control processing to be executed by the client terminal.
[0020] FIG. **12** is a flowchart illustrating an example of print job reception/transfer control processing to be executed by the CPS.
[0021] FIG. **13** is a flowchart illustrating an example of print control processing to be executed by the printing apparatus.
[0022] FIGS. **14**A, **14**B, and **14**C illustrate examples of transmission/reception packets indicating a

request for capability information and a response.
[0023] FIG. **15** is a sequence diagram illustrating an example of printer registration processing and print setting extension processing according to a second exemplary embodiment.
[0024] FIG. **16** illustrates an example of a print setting screen to be displayed by a print extension application according to the second exemplary embodiment.
[0025] FIGS. **17**A and **17**B illustrate examples of attribute information corresponding to a print job.
[0026] FIG. **18** is a flowchart illustrating an example of control processing to be executed by a client terminal according to the second exemplary embodiment.
DESCRIPTION OF THE EMBODIMENTS
[0027] Exemplary embodiments of the present disclosure will be described below.
[0028] First, the premises will be described. In recent years, more and more organizations and companies are choosing to use client terminals of a plurality of operating system (OS) vendors (e.g., Windows® OS, macOS®, or Chrome® OS). In this case, capabilities that are supported by each print client may vary depending on the type of the client terminal. For example, a print client for an OS having a large share in the market is frequently updated and thus the capabilities supported by the client are improved, while a client for an OS having a small share in the market is updated less often and thus the capabilities supported by the client may be at a low level. In addition, for example, a print client for a client terminal, such as a mobile terminal, which is not often required to make complicated print settings, may be configured to intentionally lower the support capabilities such that a user can intuitively make settings.
[0029] It is also generally known that when capability information about a printing apparatus is acquired, unsupported capability information is discarded. Accordingly, if registration processing is simply performed during registration of a printing apparatus on a cloud print service (hereinafter also referred to as “CPS”), capability information that is not supported by the CPS is discarded and then the printer is registered.
[0030] Each print client can be improved by upgrading or the like and can additionally support a print attribute that has not been supported. However, if capabilities that are not supported by the CPS are discarded during registration of a printer on the CPS and then the printer is registered, even print settings corresponding to print attributes that are supported by both the printing apparatus and the updated client cannot be made in some cases. In addition, every time a print attribute supported by the print client is added, an updating operation, such as an operation for expanding the print attributes supported by the CPS and reregistering the printing apparatus on the CPS, may need to be performed. This operation requires an excessive amount of time and labor for an administrator or the like, and thus this operation is impractical. Further, it is also possible to construct a mechanism for setting items in addition to standard items by expanding print settings provided by a print client using a print extension application. Even in such a case, if attributes that are not supported by the CPS are discarded, items other than the standard items cannot be acquired from the CPS.
[0031] In view of the above-described premises, an exemplary embodiment of the present disclosure provides a mechanism in which capability information that is not supported by a cloud print service is registered during registration of a printing apparatus on the cloud print service, and a notification indicating the capabilities of the print apparatus is appropriately sent to a client. This mechanism will be described in detail below with reference to the drawings.
[0032] The following exemplary embodiments are not intended to limit the disclosure, and not all combinations of features described in the exemplary embodiments are essential to the solving means of the disclosure.
[0033] A configuration of a printing system according to a first exemplary embodiment will now be described with reference to FIG. **1**. The printing system according to the present exemplary embodiment includes a printing apparatus **101**, client terminals **103** to **105**, and a cloud print service (hereinafter also referred to as “CPS”) **102**. The printing apparatus **101** communicates with

the CPS **102** on the Internet via a network **100**. The network **100** may be configured using a combination of a communication network, such as a local area network (LAN) or a wide area network (WAN), a cellular network (e.g., Long Term Evolution (LTE) or fifth generation mobile communication system (5G)), a wireless network conforming to Institute of Electrical and Electronics Engineers (IEEE) 802.11, and the like. In other words, the network **100** may have any configuration as long as data reception and transmission can be performed via the network **100** and may adopt any communication method for physical layers. The client terminals **103** to **105** also communicate with the CPS **102** on the Internet via a communication network or a cellular network.
[0034] The printing apparatus **101** includes a scanning function for transmitting data based on an image obtained by scanning information using a scanner to the outside of the printing apparatus **101**, a printing function for printing the image on a sheet, such as paper, based on a print job received from an external apparatus, and a copying function. Further, the printing apparatus **101** can receive a print job via the CPS **102** and execute printing. The present exemplary embodiment illustrates an example where a multi-function peripheral (MFP) including a plurality of functions is used as an example of the printing apparatus **101**. However, the present exemplary embodiment is not limited to this example. For example, a single function peripheral (SFP) including only the printing function may be used. Further, the present exemplary embodiment illustrates an example where printing is executed on a sheet, such as paper. However, the present exemplary embodiment is not limited to this example. The present exemplary embodiment can also be applied to printing control processing in, for example, three-dimensional (3D) printing for forming a three-dimensional object based on three-dimensional shape data.
[0035] The CPS **102** receives a print job from a client terminal, such as the client terminal **103**, **104** or **105**, and stores the print job. Then, the CPS **102** sends a notification that the print job is input to the printing apparatus **101** that is registered on the CPS **102**. The printing apparatus **101** that has received the notification acquires the print job and executes print processing.
<Description of Cloud Print Service of Related Art>
[0036] Next, printer registration and printing in a generally known CPS will be described with reference to FIGS. **2** and **3**. FIG. **2** is a sequence diagram illustrating an example of printer registration and printing sequences in the general CPS. FIG. **3** illustrates an example of a print setting screen to be displayed when print data is input to the CPS from the client terminal.
[0037] First, in steps S**201** and S**202**, a printing apparatus is registered on the CPS based on an operation performed by a user such as an administrator. When the processing of steps S**201** to S**202** is executed and a cloud printer is registered on the CPS, then in step S**203**, the printing apparatus notifies the CPS of attribute information indicating printing capabilities. The printing apparatus that has received the attribute information in step S**203** analyzes the attribute information. Specifically, in step S**204**, the CPS discards a print attribute that is not supported by the CPS and extracts a print attribute that is supported by the CPS. Then in step S**205**, the CPS associates the supported print attribute with the registered printer. The processing of steps S**201** to S**204** enables registration of an output destination printer having capabilities equivalent to those of the printing apparatus on the CPS. The present exemplary embodiment assumes a case where the CPS functions as an Internet Printing Protocol (IPP) proxy for relaying the print job received from the client terminal to the printing apparatus. In this case, attribute information that is not supported by the CPS (incomprehensible attribute information) is discarded and printing via the cloud is implemented by using only supported attribute information in accordance with the IPP rules.
[0038] Next, in step S**206**, the CPS sends a notification indicating the capability update result. Step S**206** illustrates a case where attribute information that is supported by the printing apparatus but is not supported by the CPS is present. In this case, a status code (successful-ok-but-ignored-attributes) indicating that a part of the attribute information is ignored is sent as a response.
[0039] Next, print processing to be executed from the client terminal will be described. Upon detecting that a printer search instruction is issued by the user, in steps S**207** to S**209**, the client

terminal searches for peripheral printers and printers on the CPS and displays a list of printers that can be used by the user. The cloud printer search is performed as follows. For example, the client terminal transmits account information about the user who has logged in to the client terminal to the CPS. The CPS extracts output destination printers that can be used by the user and are managed on the CPS. Then, in step S**208**, the CPS notifies the client terminal of the extracted output destination printers. In step S**209**, the client terminal that has received the notification displays a list of the printers based on the notification.

[0040] Next, in step S**210**, the client terminal receives a user operation for selecting a printer. In a case where the selected printer is one of the printers managed on the CPS and the printer is not registered in the client terminal, then in steps S**211** and S**212**, processing for registering the output destination printer on the client terminal is performed. In the registration processing, processing for registering the output destination printer for executing printing via the CPS is executed on a print framework provided by the OS of the client terminal. For example, when Windows® is used as the OS, print queue generation processing and printer driver configuration processing are performed.

[0041] Then, in step S**213**, the client terminal displays the print setting screen corresponding to the registered output destination printer, and receives a change of print settings performed by a user operation. Upon detecting a user operation for starting printing, in step S**214**, the client terminal generates a print job based on a printing target file and print settings, and transmits the print job to the CPS.

[0042] In steps S**215** to S**217**, the CPS discards unsupported attribute information from the attribute information indicating the print settings included in the print job, and stores the print job including supported attribute information and print data included in the job in a storage area. Then, in step S**218**, the CPS transmits, to the printing apparatus, an event (notification) indicating that the print job is input. The notification may be directly sent to the printing apparatus from the CPS, or may be made in such a manner that the printing apparatus periodically makes an inquiry about an event occurrence status and the CPS sends an event as a response to the inquiry.

[0043] In step S**219**, the printing apparatus that has received the event transmits a request for acquiring the print job to the CPS. In step S**220**, the CPS that has received the request transmits the print job including the supported attribute information to the printing apparatus that has transmitted the request. Lastly, in step S**221**, the printing apparatus that has received the print job prints an image on a sheet based on the print job, and thus a series of processing is terminated.

[0044] In the processing of the related art described above with reference to FIG. **2**, when capability information about the printing apparatus is acquired, unsupported capability information is discarded. Accordingly, in the case of registering the printing apparatus on the CPS, if the registration processing is simply performed, it is considered that the printer is registered while capability information that is not supported by the CPS is discarded.

[0045] FIG. **4** is a table illustrating capabilities that are supported by the client terminals **103** to **105**, the CPS **102**, and the printing apparatus **101** according to the present exemplary embodiment. For convenience of illustration, some of the capabilities are extracted and displayed. In practice, not only these capabilities, but also capabilities indicating sheet sizes that are supported by the printing apparatus **101** and information indicating the type of sheets supported by the printing apparatus **101** are included. In addition, for example, capabilities indicating the sizes of data that can be received are also included.

[0046] The printing apparatus **101** supports all attributes listed in FIG. **4**. The present exemplary embodiment assumes a case where the printing apparatus **101** supports standard attributes defined in standards, such as Request For Comments (RFC), and vendor unique attributes that are not defined in the standards. FIG. **4** illustrates attribute information based on the IPP as examples of standard attributes.

[0047] The attribute information will be described in detail. A “print-color-mode-supported” attribute is attribute information about a color mode. This attribute can take three types of values,

i.e., “auto”, “color”, and “monochrome”. The corresponding print setting is referred to as a “color mode” setting, and this setting can take values of “auto”, “color”, and “monochrome”. The “color mode” is a setting for selecting a color setting for print output. When “color” is selected, a color output is performed. When “monochrome” is selected, a monochrome output is performed. When “auto” is selected, the printing apparatus **101** performs a color output or a monochrome output.
[0048] A “print-quality-supported” attribute is attribute information about a print quality. This attribute can take three types of values, i.e., “high”, “normal”, and “draft”. The corresponding print setting is referred to as a “print quality” setting, and this setting can take values of “high”, “normal”, and “draft”. The “print quality” setting is a setting for selecting the amount of toner to be consumed. When “high” is selected, printing is executed with a large amount of toner to be consumed. When “normal” is selected, printing is executed with a normal amount of toner to be consumed. When “draft” is selected, printing is executed with a less amount of toner to be consumed.
[0049] A “number-up-supported” attribute is attribute information about N-up printing. This attribute can take numerical values, and indicates that pages corresponding to a designated value can be collectively printed on one printing surface of a sheet. For example, when values “1, 2, 4” are stored, “1-in-1”, “2-in-1”, and “4-in-1” can be set in the print setting.
[0050] A “finishing-supported” attribute is attribute information about post-processing. This attribute can take numerical values, and values corresponding to finishing information that is supported by the printing apparatus **101** are stored. For example, “20” is associated with “staple-top-left” and “22” is associated with “staple-top-right”. These values are treated as “staple” setting values in the print setting, and enable setting of, for example, “stapling at an upper left position” and “stapling at an upper right position”.
[0051] An “emphasize-thin-lines-supported” attribute is an example of vendor unique attributes, and is attribution information about how to treat thin lines. This attribute can take values of “true” and “false”. The print setting corresponding to the attribute information is referred to as an “emphasize thin lines” setting. When “true” is selected, printing in which thin lines are emphasized is executed. When “false” is selected, printing in which thin lines are not emphasized is executed. This setting is used to, for example, prevent a phenomenon in which some of thin lines are not fully reproduced or are discontinuously formed in an inexpensive printing apparatus with a low printing resolution or the like.
[0052] Similarly, a “color-adjust-supported” attribute is a vendor unique attribute, and is a setting for special printing to execute printing by strengthening a specific color. This attribute can take values of “strengthen blue”, “strengthen yellow”, and “strengthen red”. The corresponding print setting is referred to as a “color adjust” setting. In this setting, “none”, “strengthen blue”, “strengthen yellow”, and “strengthen red” can be set.
[0053] The printing apparatus **101** supports all the pieces of attribute information described above. However, at this time, the CPS **102** does not support an attribute corresponding to a function for stapling documents at two positions. On the other hand, a print client installed in the client terminal **103** is improved by upgrading and additionally supports the print attribute (e.g., the function for stapling documents at two positions) that has not been supported.
[0054] In view of the above-described circumstances, in the case of registering a printer on the CPS, if control processing for simply registering the printer while discarding capabilities that are not supported by the CPS is performed, the following issue is raised. That is, the print setting corresponding to the print attribute that is actually supported by both the printing apparatus and the updated client cannot be made in printing via the CPS.
[0055] A specific example will be described with reference to FIG. **3**. FIG. **3** illustrates a screen to be displayed when print settings are made by the print client having capabilities equivalent to those of the client terminal **103**. When the simple registration processing is performed like in the related art, in the registration processing of steps S**203** to S**205**, the output destination printer is registered

on the CPS in a state where an attribute value for stapling documents at two positions and attribute information about N-up printing are discarded. The print client acquires the capability information obtained by performing rounding processing on the CPS in the processing of steps S**211** and S**212**. This may cause an issue that the setting for N-up printing and complicated stapling settings that are actually supported by both the printing apparatus and the client cannot be made via the print setting screen illustrated in FIG. **3**. In addition, every time a print attribute supported by the print client is added, the print attributes supported by the CPS are also expanded, and an updating operation, such as an operation for reregistering the printing apparatus on the CPS, may need to be performed. This operation requires an extra amount of time and labor for the administrator, and thus this operation is impractical.

[0056] In view of the above-described issues, in the present exemplary embodiment, control processing is executed such that, in the case of registering the printing apparatus on the CPS, capability information that is not supported by the CPS is also registered as capability information about the printing apparatus and a notification indicating the capabilities of the printing apparatus is appropriately sent to the client. This processing will be described in detail below.

<Hardware Configuration of Printing Apparatus **101**>

[0057] First, the printing apparatus **101** will be described with reference to FIG. **5**. FIG. **2** is a block diagram illustrating a hardware configuration of the printing apparatus **101**. The printing apparatus **101** includes a scanning function for scanning an image on a sheet, and a file transmission function for transmitting the scanned image to an external communication apparatus. The printing apparatus **101** also includes a printing function for printing an image on a sheet. The printing apparatus **101** also includes a function for receiving a print job from the CPS **102** and executing printing based on the received print job.

[0058] A control unit **110** including a central processing unit (CPU) **111** controls the overall operation of the printing apparatus **101**. The CPU **111** reads out control programs stored in a read-only memory (ROM) **112** or a storage **114**, and performs various control processing such as printing control and scanning control. The ROM **112** stores control programs that can be executed by the CPU **111**. A random access memory (RAM) **113** is a main storage memory to be accessed from the CPU **111**, and is used as a work area or a temporary storage area for loading various control programs. The storage **114** stores print data, image data, various programs, and various setting information. Thus, the hardware modules, such as the CPU **111**, the ROM **112**, the RAM **113**, and the storage **114**, constitute a computer.

[0059] The printing apparatus **101** according to the present exemplary embodiment is configured to execute each process in flowcharts to be described below in such a manner that a single CPU **111** uses a single memory (RAM **113**). However, the printing apparatus **101** may have any configuration other than the above-described configuration. For example, the processes in the flowcharts to be described below can also be executed by causing a plurality of processors, memories, and storages to operate in cooperation. Some of the processes may be executed using a hardware circuit.

[0060] A printer interface (I/F) **119** connects a printer **120** (printer engine) and the control unit **110**. The printer **120** prints an image on a sheet fed from a sheet feeding cassette (not illustrated) based on print data input via the printer I/F **119**. As a printing method, an electrophotographic method for transferring toner onto a sheet and fixing the toner onto the sheet, or an inkjet method for discharging ink onto a sheet to execute printing may be used.

[0061] A scanner I/F **117** connects a scanner **118** and the control unit **110**. The scanner **118** scans a document placed on a platen glass (not illustrated) and generates image data. The image data generated by the scanner **118** is printed by the printer **120**, is stored in the storage **114**, or is transmitted to an external apparatus via a network I/F **121**.

[0062] An operation unit I/F **115** connects an operation unit **116** and the control unit **110**. The operation unit **116** is provided with a liquid crystal display unit including a touch panel function and various hardware keys. The operation unit **116** functions as a display unit that displays information for the user, and also functions as a reception unit that receives an instruction from the user. The CPU **111** operates in cooperation with the operation unit **116** to perform information display control processing and control processing for receiving a user operation.

[0063] The network I/F **121** is connected to a network cable and thus can execute communication with an external apparatus on the network **100** or the Internet. The present exemplary embodiment assumes a case where the network I/F **121** is a communication interface for establishing a wired communication conforming to Ethernet®. However, the present exemplary embodiment is not limited to this case. For example, the network I/F **121** may be a wireless communication interface conforming to IEEE 802.11 series. Alternatively the network I/F **121** may be a communication interface for establishing both of a wired communication and a wireless communication. Further, a communication interface for establishing a mobile communication, including a 3G line such as Code Division Multiple Access (CDMA), a 4G line such as LTE, and 5G New Radio (NR), may also be used.

<Hardware Configuration of Server Resource for Providing CPS **102**>

[0064] FIG. **6** is a block diagram illustrating a hardware configuration of a server as a real resource for providing the CPS **102** illustrated in FIG. **1**. The server that provides the CPS **102** is hereinafter also referred to as the server **102**. A CPU **131** operates in cooperation with each unit to control the operation of the server **102**. The CPU **131** reads out an OS and control programs stored in a ROM **132** or a storage **134**, and executes the OS and control programs. The ROM **132** stores control programs that can be executed by the CPU **131**. A RAM **133** is a main storage memory for the CPU **131**, and is used as a work area or a temporary storage area for loading various control programs. The storage **134** stores print data, image data, various programs, and various setting information. The present exemplary embodiment assumes a case where an auxiliary storage device, such as a hard disk drive (HDD), is used as the storage **134**. However, a nonvolatile memory, such as a solid state drive (SSD), may be used instead of the HDD. Thus, the hardware modules, such as the CPU **131**, the ROM **132**, and the RAM **133**, constitute a computer. The server **102** may further include an application specific integrated circuit (ASIC) for rendering print data. The present exemplary embodiment illustrates a case where the processes in the flowcharts to be described below are executed in such a manner that a single CPU **131** uses a single memory (RAM **133**), for ease of explanation. However, the present exemplary embodiment may have any configuration other than the above-described configuration. For example, the processes in the flowcharts to be described below can also be executed by causing a plurality of processors, RAMs, ROMs, and storages to operate in cooperation. The processes can also be executed using a plurality of server computers. The server **102** can provide a plurality of different tenants with the CPS by using a containerization or virtualization technique.

[0065] A network I/F (interface) **137** is an interface for communicating with an external apparatus via a network. The server **102** is connected to the Internet via the network I/F **137**. An operation unit I/F **135** is an interface for connecting an operation unit **136** such as a keyboard, a mouse, and a display. Input/output devices connected to the operation unit I/F **135** are used for, for example, maintenance of the real server for providing the CPS.

[0066] The hardware configuration of each of the client terminals **103** to **105** is similar to that of the server **102**.

[0067] The CPS **102** provided by the resource of the server **102** includes a function of registering and managing an output destination printer, a function of managing the progress of each print job, and a function of spooling print jobs. The CPS **102** stores output destination printer information for managing the output destination printer. The output destination printer information is management information in which attribute information indicating the capabilities of the printing apparatus, information used for communication with the printer (e.g., an internet protocol (IP) address, a printer host name, and an access token required for communication), and information for identifying the model name of the printing apparatus are associated. The management information is referred to, as needed, in the flowcharts to be described below.

[0068] Further, Windows®, Android®, macOS®, iOS®, or the like is installed in each of the client terminals **103** to **105**. The user can use the print client and the print framework for the CPS **102** that are preliminarily installed in the OS. The print client and the print framework provide the user with a status monitor function for monitoring the progress of printer search processing, registration processing, print setting screen display processing, print data generation processing, processing for communication with the CPS **102**, and printing. The present exemplary embodiment illustrates a case where the print client is preliminarily installed. However, the present exemplary embodiment is not limited to this case. For example, the print client may be subsequently installed by the user.

Printer Registration and Printing Sequences in First Exemplary Embodiment

[0069] Next, the printer registration on the CPS **102** and printing processing according to the first exemplary embodiment will be described with reference to FIG. **7** and FIGS. **14**A to **14**C. FIG. **7** is a sequence diagram illustrating an example of a processing sequence. FIGS. **14**A to **14**C illustrate examples of exchange of capability information.

[0070] Steps S**701** and S**702** are registration processing similar to that of steps S**201** and S**202** illustrated in FIG. **2**. The user, such as the administrator, selects a registration destination CPS via the operation unit **116** of the printing apparatus **101**, and selects a "registration" button. The printing apparatus **101** that has detected the selection starts a registration operation. In step S**701**, the printing apparatus **101** that has detected the selection transmits a registration processing request to the CPS **102**. A protocol for registration processing conforms to the protocol defined in the CPS. The present exemplary embodiment is described assuming that, for example, Register-Output-Device operation defined in Printer Working Group (PWG) 5100.22 is used. The CPS **102** that has received the registration processing request checks the content of the registration request. In step S**702**, the CPS **102** transmits a registration processing request response to the printing apparatus **101**. If the content of the registration processing request is normal, a successful response is sent. If the content of the registration processing request is not normal, an unsuccessful response is sent. The present exemplary embodiment is described assuming that the successful response is sent. In the processing of steps S**701** and S**702**, output destination printer information is registered on the CPS **102**. Further, in the registration processing, an access token for accessing the CPS **102** from the printing apparatus **101** is issued. The printing apparatus **101** stores the access token in the storage **114** and uses the access token, for example, when the printing apparatus **101** sends an inquiry to the CPS **102** or makes a request.

[0071] Next, in step S**703**, the printing apparatus **101** transmits a capability notification including capability information supported by the printing apparatus **101** to the CPS **102**. The present exemplary embodiment assumes a case where, for example, Update-Output-Device-Attributes operation defined in PWG 5100.18 is used. In the processing of step S**703**, for example, a packet indicating capability information illustrated in FIG. **14**A is transmitted. The packet illustrated in FIG. **14**A includes a list of character strings of attribute information illustrated in FIG. **4**. In other words, the packet illustrated in FIG. **14**A includes standard attribute information conforming to the IPP and vendor unique attribute information.

[0072] Next, in step S**704**, the CPS **102** that has received the capability information from the printing apparatus **101** registers not only the attribute information supported by the CPS **102** but also unsupported attribute information as output destination printer information. The supported attribute information and the unsupported attribute information are stored in such a manner that the supported attribute information and the unsupported attribute information can be distinguished from each other.

[0073] Then, in step S**705**, the CPS **102** transmits "Success-ok", which is a registration successful response illustrated in FIG. **14**C to the printing apparatus **101**. A status message is added to the information as additional information. The status message includes information indicating that some attributes are not supported by the CPS, or that an unsupported attribute is registered. This processing enables the printing apparatus registration processing using specification information that also includes information other than the attributes supported by the CPS and enables storage of the original specification information about the printing apparatus.

[0074] Next, a sequence for executing printing via the CPS from the client terminal (e.g., the client terminal **103**) will be described with reference to steps S**706** to S**712**. First, the user executes printer search processing on the CPS **102** from the operation unit of the client terminal **103**. In step S**706**, the client terminal **103** that has detected a print search instruction from the user transmits a printer search request to the CPS **102**. The search request includes cloud account information corresponding to the user who has logged in to the client terminal **103**. The cloud account information is used to identify the tenant to which the user belongs. The printer search processing may be performed by a method of searching for printers by designating all printers, which are registered on the CPS **102** and managed by the tenant to which the user belongs, or by a method of searching for printers using filter conditions such as a specific printer name or location.

[0075] In step S**707**, the CPS **102** that has received the search request from the client terminal **103** transmits printer information that indicates the printer registered on the CPS **102** and matches the search request to the client terminal **103**. At this time, there is no need to send a response including specification information associated with the printer, and it only needs to send a response including only representative attribute information, such as the printer name, which is required for selecting the output destination. In step S**708**, the client terminal **103** that has received the printer list information from the CPS **102** displays a list of the printers on the operation unit. The client terminal **103** may search for printers on a local network by using a search protocol, such as a multicast Domain Name System (mDNS) protocol, and may merge the search results and display the search results on a printer list screen. It is assumed that the output destination printers registered on the print framework of the client terminal **103** are also displayed in a list on the screen. In step S**709**, the user selects a desired printer from the printer list. The present exemplary embodiment illustrates an example where the printer that is registered on the CPS **102** but is not registered in the client terminal **103** is selected.

[0076] In step S**710**, the client terminal **103** that has detected that the printer is selected transmits a printer capability acquisition request to the CPS **102** so as to acquire the detailed specification of the printer. The present exemplary embodiment assumes a case where the printer capability acquisition request is made by, for example, Get-Printer-Attributes operation defined in the IPP. In step S**711**, the CPS **102** that has received the printer capability acquisition request transmits a printer capability response including the attribute information supported by the CPS **102** and unsupported attribute information to the client terminal **103**. In the processing of step S**711**, for example, a packet indicating capability information illustrated in FIG. **14**B is transmitted. The packet illustrated in FIG. **14**B includes a list of character strings of attribute information registered on the CPS **102** in step S**704**. The attribute information including attributes that are not supported by the CPS **102** but are supported by the printing apparatus **101** as illustrated in FIG. **14**B can be notified. Then, in step S**712***a*, the client terminal **103** extracts attribute information supported by the print client in the terminal, and registers output destination printer information by associating the extracted attribute information with the output destination. In the processing of step S**712***a*, the unsupported attribute information that is not supported by the print client is discarded.

[0077] Next, step S**712***b*, the client terminal **103** displays a print setting screen based on the output destination printer information registered in step S**712***a*. FIG. **8** illustrates an example of the print setting screen to be displayed in step S**712***b*. As described above with reference to FIG. **4**, the print client of the client terminal **103** supports the attribute values for “number-up” printing and stapling output materials at two positions, and thus these setting values can be appropriately displayed. In FIG. **8**, portions in bold letters indicate print settings that cannot be displayed in the print setting screen of the related art described above with reference to FIG. **2**. Accordingly, a printing user interface (UI) can be appropriately displayed even when the print client of the client terminal is upgraded and the print client supports new functions. This therefore increases the possibility that the user can make desired print settings.

[0078] Next, a sequence to be performed when printing is executed will be described with reference to steps S**713** to S**719**. In step S**713**, the user changes the print settings via the screen illustrated in FIG. **8** as needed, and presses a print button. In step S**714**, the client terminal **103** that has detected that the print button is pressed transmits a print request including attribute information indicated by the print settings to the CPS **102**. The present exemplary embodiment assumes a case where the print request is made by Create-Job operation or Print-Job operation defined in the IPP. When the instruction is made by Print-Job operation, the attribute information and print data are transmitted. When the instruction is made by Create-Job operation, a plurality of pieces of document data is registered as a printing target by using an operation for adding a subsequent printing target document.

[0079] Next, in step S**715**, the CPS **102** analyzes the print execution request received from the client terminal **103** and stores print attributes (information indicating print settings) included in the request and information about the print job associated with document data to be printed in the storage of the CPS **102**.

[0080] Also, in this case, even if an unsupported attribute value is included in the request, the CPS **102** stores the unsupported attribute in a similar manner to the supported attributes. During the storage, processing, such as rendering of print data, may be performed on the CPS **102**.

[0081] Next, in step S**716**, the CPS **102** sends a notification that the print job is input to the printing apparatus **101**. For the notification, for example, Get-Notification operation defined in the IPP can be used.

[0082] In step S**717**, the printing apparatus **101** that has detected that the print job is present in the CPS **102** sends a print job acquisition request to the CPS **102**. For the print job acquisition request, for example, Fetch-Job operation defined in the IPP can be used. In step S**718**, the CPS **102** that has received the print job acquisition request sends the print data and print attribute values stored in the storage as a response. In the present exemplary embodiment, the unsupported attribute information is not discarded but is stored in step S**715**. Thus, the attribute information that is not supported by the CPS **102** can also be appropriately transmitted to the printing apparatus **101**. In step S**719**, the printing apparatus **101** prints an image on a sheet based on the received print job.

[0083] For example, in a case where the print setting “2-in-1” is made on the screen of FIG. **8** that is displayed on the client terminal **103**, the attribute information corresponding to the setting “2-in-1” is not discarded by the CPS **102** but is transmitted to the printing apparatus **101**. Accordingly, the printing apparatus **101** can determine that N-up printing is to be performed based on the attribute information, and thus can appropriately perform N-up printing.

[0084] Control processing in each device constituting the printing system will be described with reference to flowcharts illustrated in FIGS. **9** to **13**. FIGS. **9** and **13** illustrate processing in the printing apparatus **101**. FIGS. **10** and **12** illustrate processing in the CPS **102**. FIG. **11** illustrates processing in the client terminal.

[0085] Each operation (step) in the flowchart illustrated in FIG. **9** is implemented by the CPU **111** executing control programs stored in the ROM **112** or the storage **114**. The processing illustrated in FIG. **9** is executed when a registration request for registering a printer on the CPS **102** is received from the administrator.

[0086] In step S**901**, the CPU **111** transmits the registration request to the CPS **102**. In step S**902**, the CPU **111** determines whether the printer registration on the CPS **102** is successful. Specifically, if the registration result received from the CPS **102** indicates “successful”, it is determined that the registration is successful (YES in step S**902**). Then, the processing proceeds to step S**903**. On the other hand, if the registration result received from the CPS **102** indicates “unsuccessful”, or if no response is received after a lapse of a predetermined period, it is determined that the registration is unsuccessful (NO in step S**902**). Then, the processing proceeds to step S**907**.

[0087] In step S**903**, the CPU **111** transmits the capability notification illustrated in FIG. **14**A to the CPS **102**. In step S**904**, the CPU **111** determines the type of the response to the capability notification transmitted in step S**903**. For example, when the response indicates a format error, an authentication error, or the like, the CPU **111** determines the type of the response to be an error response (ERROR RESPONSE in step S**904**). Then, the processing proceeds to step S**907**. On the other hand, when the response is a successful response, such as "Success-ok" as illustrated in FIG. **14**C, the CPU **111** determines the type of the response to be a successful response (SUCCESSFUL RESPONSE in step S**904**). Then, the processing proceeds to step S**906**. When the response is a response indicating that the registration is successful but some of attributes are discarded, such as "Successful-ok-but-ignored-Attributes", the CPU **111** determines the type of the response to be a conditional successful response (CONDITIONAL SUCCESSFUL RESPONSE in step S**904**). Then, the processing proceeds to step S**905**.

[0088] In step S**905**, the CPU **111** displays the registration result indicating that the registration is successful but some of the attributes are ignored on the operation unit **116**. In step S**906**, the CPU **111** displays the registration result indicating that the registration is successful on the operation unit **116**. In step S**907**, the CPU **111** displays the registration result indicating that the registration is unsuccessful on the operation unit **116**.

[0089] Next, output destination printer registration processing on the CPS that is executed by the server **102** that provides the CPS **102** will be described with reference to FIG. **10**. Each operation (step) in the flowchart illustrated in FIG. **10** is implemented by the CPU **131** included in the server **102** when the CPU **131** executes control programs stored in the ROM **132** or the storage **134**. As described above with reference to FIG. **6**, the processing can also be executed by causing a plurality of computer resources to operate in cooperation. Further, communication processing are implemented by an operation in cooperation with the network I/F **137**.

[0090] In step S**1001**, the CPU **131** of the server **102** receives a registration request from the printing apparatus **101**, and executes registration processing for registering the output destination printer on the CPS **102**. The CPU **131** creates output destination printer information for managing the cloud printer on the CPS **102**. Further, an access token for accessing the CPS **102** from the printing apparatus **101** is issued. After completion of the registration processing, the processing proceeds to step S**1002**. In step S**1002**, the CPU **131** sends the registration processing result as a response to the printing apparatus **101**. In a case where the registration processing is unsuccessfully completed, for example, when the user authentication is unsuccessful, or when the request content is not normal, a notification indicating an error is sent. When the error notification is sent, the CPU **131** interrupts the registration processing without performing subsequent capability update processing.

[0091] Next, in step S**1003**, the CPU **131** receives the capability notification illustrated in FIG. **14**A from the printing apparatus **101** that has been successfully registered in step S**1002**. In step S**1004**, the CPU **131** performs analysis processing on the attribute information included in the capability notification received in step S**1003**. Specifically, the CPU **131** checks if the received content satisfies a prescribed format, or if the received content includes data used for an injection attack or the like. If the received content does not satisfy the format, or if the received content includes data to be used for an injection attack or the like, it is determined that the content is not appropriate, and information indicating an error is recorded as the analysis result. On the other hand, if the content is normal, information indicating that the content is normal is recorded as the analysis result. Further, the CPU **131** performs processing for classifying the attribute information into attribute information that is supported by the CPS **102** and attribute information that is not supported by the CPS **102**. When at least one piece of attribute information that is not supported by the CPS **102** is included, the CPU **131** records the analysis result indicating that the unsupported attribute is included. On the other hand, when the attribute information that is not supported by the CPS **102** is not included, the CPU **131** records the analysis result indicating that only the supported attribute information is included.

[0092] In step S**1005**, the CPU **131** determines the type of the result of the analysis processing executed in step S**1004**. If the analysis result indicates an error (ERROR in step S**1005**), the processing proceeds to step S**1006**. If the analysis result indicates that the content is normal and only the supported attribute is included (CONTENT IS NORMAL AND ONLY SUPPORTED ATTRIBUTE IS INCLUDED in step S**1005**), the processing proceeds to step S**1007**. On the other hand, if the analysis result indicates that the content is normal and the unsupported attribute is included (CONTENT IS NORMAL AND UNSUPPORTED ATTRIBUTE IS INCLUDED in step S**1005**), the processing proceeds to step S**1009**.

[0093] In step S**1006**, the CPU **131** deletes the output destination printer information registered in step S**1001**, transmits an error response to the printing apparatus **101**, and terminates the series of registration processing.

[0094] In step S**1007**, the CPU **131** updates information indicating the printing capabilities included in the output destination printer information registered in step S**1001** with the attribute information acquired in step S**1003**. In other words, the printing capabilities of the printing apparatus **101** are registered in the output destination printer information. Next, in step S**1008**, the CPU **131** transmits a normal response to the printing apparatus **101**.

[0095] In step S**1009**, the CPU **131** updates the information indicating the printing capabilities included in the output destination printer information registered in step S**1001** with the attribute information classified in step S**1005**. Specifically, the attribute information that is supported by the CPS **102** and the attribute information that is not supported by the CPS **102** are registered in the output destination printer information in such a manner that the supported attribute information and the unsupported attribute information can be identified. Then, in step S**1010**, the CPU **131** transmits, to the printing apparatus **101**, the normal response indicating information indicating that the unsupported attribute information is registered as described above with reference to FIG. **14**C. After completion of the response, the series of registration processing is terminated.

[0096] The processing described above with reference to FIG. **10** makes it possible to register the capability information corresponding to the full specification of the printing apparatus described above with reference to FIG. **4** on the CPS **102**.

[0097] Next, control processing to be performed when the printer registered on the CPS **102** is used from the client terminal will be described with reference to FIG. **11**.

[0098] Each operation (step) in the flowchart illustrated in FIG. **11** is implemented by the CPU **131** of the client terminal (e.g., the client terminal **103**) when the CPU **131** executes control programs stored in the ROM **132** or the storage **134**. The flowchart illustrated in FIG. **11** is executed when an operation for searching for printers is received from the user.

[0099] In step S**1101**, the CPU **131** of the client terminal **103** searches for printers and displays a list of found printers.

[0100] Next, in step S**1102**, the CPU **131** of the client terminal **103** determines whether a user operation for selecting a printer that is not registered in the print client has been received. If the user operation for selecting a printer that is not registered in the print client has been received (YES in step S**1102**), the processing proceeds to step S**1103**. If the user operation for selecting a printer that is registered in the print client has been received (NO in step S**1102**), the processing proceeds to step S**1109**. In step S**1109**, the CPU **131** of the client terminal **103** sets the selected registered printer as the output destination. Next, the CPU **131** refers to the output destination printer information corresponding to the selected printer, and acquires attribute information. Then, the CPU **131** generates a print setting screen based on the acquired attribute information. After the generation of the print setting screen is completed, the CPU **131** displays the generated print setting screen on the operation unit **136**, and terminates a series of processing. After completion of the series of processing, print setting change processing and print job transmission processing, which are described above with reference to FIG. **7**, are performed.

[0101] In step S**1103**, the CPU **131** of the client terminal **103** transmits an attribute information acquisition request to the selected printer, and receives a response to the request. Next, in step S**1104**, the CPU **131** analyzes the attribute information, discards the attribute information that is not supported by the client, and extracts only the attribute information that is supported by the client. Then, the output destination printer information, in which the extracted attribute information supported by the client is associated with information for identifying the output destination (e.g., an IP address, a host name, and the like of the CPS **102**), is registered on the print framework provided by the OS. For example, when Windows® is used as the OS, the registration of output destination printer information indicates processing for generating a print queue for transmitting data to the CPS **102** and configuring a printer driver based on the acquired capability information.

[0102] Next, in step S**1107**, the CPU **131** of the client terminal **103** sets the printer for which the registration processing is completed as the output destination. Then, the CPU **131** acquires attribute information corresponding to the printer set as the output destination and generates a print setting screen. After the generation of the print setting screen is completed, the CPU **131** displays the generated print setting screen on the operation unit **136**, and then terminates the series of processing. After completion of the series of processing, the print setting change processing and print job transmission processing described above with reference to FIGS. **7** and **8** are performed.

[0103] The processing described above makes it possible to appropriately display a print UI even when the print client of the client terminal is upgraded and supports new functions.

[0104] Next, control processing to be performed when the CPS **102** receives a print job from the client terminal will be described with reference to the flowchart illustrated in FIG. **12**. Each operation (step) in the flowchart illustrated in FIG. **12** is implemented by the CPU **131** included in the server **102** when the CPU **131** executes control programs stored in the ROM **132** or the storage **134**, like in FIG. **10**.

[0105] In step S**1200**, the CPU **131** of the server **102** determines whether a request for the print job has been received from the client terminal. If the request for the print job has been received from the client terminal (YES in step S**1200**), the processing proceeds to step S**1201**. If the request for the print job has not been received from the client terminal (NO in step S**1200**), the processing proceeds to step S**1205**. The present exemplary embodiment assumes a case where the request for the print job is made by Create-Job operation or Print-Job operation defined in the IPP. However, the present exemplary embodiment is not limited to this case. For example, Validate-Job operation by which a communication partner makes an inquiry about whether a print job can be processed before the client transmits the print job may be used.

[0106] In step S**1201**, the CPU **131** of the server **102** receives a series of print jobs including print data to be transmitted after the print request is received from the client terminal. Next, in step S**1202**, the CPU **131** determines whether the attribute information indicating print settings for the print jobs includes attribute information that is not supported by the CPS **102**. If the attribute information does not include attribute information that is not supported by the CPS **102** (NO in step S**1202**), the processing proceeds to step S**1203**. If the attribute information includes attribute information that is not supported by the CPS **102** (YES in step S**1202**), the processing proceeds to step S**1204**. When the print request received in step S**1200** is made by Create-Job operation or Validate-Job operation, only the attribute information is sent to the CPS **102** before print data (document data) is received. Accordingly, the determination in step S**1202** may be made before print data (document data) is received, and then the print data may be received.

[0107] In step S**1203**, if the server is to perform rendering processing or data conversion processing based on the supported attribute information, the CPU **131** executes rendering processing or data conversion processing. Further, if conversion processing is to be performed on print data, the converted print data and the print jobs including attribute information are stored in the storage area. If there is no need to perform conversion processing, the received print data and the print jobs including the attribute information are stored in the storage area.

[0108] On the other hand, in step S**1204**, the CPU **131** stores the print jobs including unsupported attribute information, supported attribute information, and print data in the storage area. The processing of step S**1204** differs from the processing of step S**1203** in that rendering processing or data conversion processing is not performed in consideration of the dependence relation between the unsupported attribute information and the supported attribute information.

[0109] Rendering processing or data conversion processing may not be performed on the CPS **102**. In this case, processing of simply storing the received print jobs may be performed instead of the processing of steps S**1202** to S**1204**.

[0110] Next, in step S**1205**, the CPU **131** determines whether the print job acquisition request has been received from the printing apparatus (e.g., the printing apparatus **101**). If it is determined that the print job acquisition request has been received (YES in step S**1205**), the processing proceeds to step S**1206**. If the print job acquisition request has not been received (NO in step S**1205**), the series of processing is terminated.

[0111] In step S**1206**, the CPU **131** transmits the print job corresponding to the acquisition request to the printing apparatus that has sent the request. After the transmission is completed, the series of processing is terminated. The reception processing and transmission processing illustrated in FIG. **12** may be repeatedly performed during a period in which the CPS **102** provides the print service.

[0112] Next, control processing to be performed when the printing apparatus **101** receives a print job from the CPS **102** and executes printing will be described with reference to the flowchart illustrated in FIG. **13**. Each operation (step) in the flowchart illustrated in FIG. **13** is implemented by the CPU **111** executing control programs stored in the ROM **112** or the storage **114**, like in FIG. **9**.

[0113] In step S**1301**, the CPU **111** receives a notification indicating that a print job to be executed is stored from the CPS **102**. Then, in step S**1302**, the CPU **111** transmits a print job acquisition request. Specifically, the printing apparatus **101** sends an inquiry about a list of available jobs to the CPS **102**. Then, the printing apparatus **101** transmits an acquisition request for acquiring data on each print job included in the list.

[0114] Next, in step S**1303**, the CPU **111** acquires the print job from the CPS **102**. In step S**1304**, the CPU **111** analyzes the attribute information included in the acquired print job, discards the attribute information that is not supported by the printing apparatus, and executes print processing. The processing described above with reference to FIGS. **9** to **13** makes it possible to make the maximum use of the capabilities supported by the client terminal **103** in a case where printing is executed via the CPS **102**. The present exemplary embodiment described above illustrates an example of the capability notification during registration of a printer for the first time. However, the present exemplary embodiment is not limited to this example. For example, the output destination printer information managed by the client terminal may be updated at predetermined intervals (e.g., every 30 days).

[0115] A second exemplary embodiment provides a mechanism for installing a print extension application provided by the vendor of the printing apparatus **101** in the client terminal **104**, in addition to the processing according to the first exemplary embodiment. This print extension application extends the appearance of print settings and a user interface for print settings on the client terminal **104**.

[0116] A hardware configuration of each device according to the second exemplary embodiment is similar to that of the first exemplary embodiment. Differences between the second exemplary embodiment and the first exemplary embodiment will be described below. The printing system according to the second exemplary embodiment includes not only the devices according to the first exemplary embodiment but also a server that distributes the print extension application. The present exemplary embodiment illustrates an example where the print extension application is installed in the client terminal **104**.

[0117] A distribution mechanism and a print extension application calling method will be described below with reference to a sequence diagram illustrated in FIG. **15**. FIG. **15** is a sequence diagram illustrating processing according to the second exemplary embodiment to be executed in place of the sequence illustrated in FIG. **7** according to the first exemplary embodiment.

[0118] The printer registration processing on the CPS **102** and the printer selection processing and capability acquisition processing in the client terminal described in steps S**1501** to S**1511** are similar to steps S**701** to S**711** illustrated in FIG. **7**, and thus the descriptions thereof are omitted.

[0119] In step S**1512**, the client terminal **104** identifies the model of the printing apparatus selected in step S**1509**. Next, information indicating the identified model (also referred to as model information) is transmitted to the distribution server, and an inquiry about whether the print extension application is present is made. In step S**1513**, the distribution server that has received the inquiry determines whether the print extension application for the model is present, and sends a response indicating whether the print extension application is present to the client terminal **104**. If the print extension application is present, the distribution server includes information about an application downloading destination in the response.

[0120] A case where the print extension application is provided from the distribution server will now be described with reference to steps S**1514** to S**1518**. In step S**1514**, the OS of the client terminal **104** downloads the print extension application from the distribution server based on the downloading destination information included in the response and installs the print extension application in the client terminal **104**.

[0121] Next, in step S**1515**, the OS of the client terminal **104** registers output destination printer information in which the attribute information including attribute information supported by the client and attribute information that is not supported by the client is associated with information for identifying the output destination.

[0122] Next, in step S**1516**, the OS of the client terminal **104** registers the correspondence relation between the print extension application and the output destination printer information in a registry database (DB). The storage destination of the correspondence relation is merely an example. The correspondence relation may be stored in any storage area that can be referred to by the OS of the client terminal **104**.

[0123] Then, in step S**1517**, the user presses an advanced print settings button through a print setting screen provided by the OS or an easy print setting screen provided by a general-purpose application. The OS of the client terminal **104** that has detected that the advanced settings button is pressed identifies information about the printer selected as the output destination when the advanced settings button is pressed. Next, in step S**1518**, the OS of the client terminal **104** refers to the registry DB based on the identified information, and identifies and starts the corresponding print extension application.

[0124] In step S**1519**, the print extension application started by the OS acquires attribute information included in the associated output destination printer information, and displays an extension print setting screen based on the acquired attribute information. FIG. **16** illustrates an example of the extension print setting screen provided by the print extension application. For example, the print extension application is an application provided by the vendor that manufactures the printing apparatus **101** and is configured such that print settings dedicated to the printing apparatus **101** are available. FIG. **16** illustrates a main screen, and each of numbers **1** to **6** in circles represents a key for shifting to the advanced settings screen. The user can make post-processing setting, such as stapling, by selecting any one of the keys. It is also possible to make print settings corresponding to vendor unique attribute information, such as color adjustment and thin line emphasizing functions, which are not supported by the general-purpose print client in the first exemplary embodiment. Upon detecting that an OK button is selected, the print extension application updates the content of the attribute information based on the settings made via the screen, and then the application is terminated. FIGS. **17**A and **17**B are diagrams illustrating print

attribute information according to the second exemplary embodiment. FIG. **17**A illustrates a packet as an example of a request for a print job to be transmitted to the CPS **102**. In this case, the packet indicates a case where thin line printing is valid and “strengthen blue” is selected as the color strengthening setting through a screen provided by the print extension application. The print client also transmits attribute information that is not supported by the print client to the CPS **102** without discarding the attribute information. The print job is temporarily stored in the CPS **102**. The CPS **102** transmits the print job to the printing apparatus **101** when the job acquisition request is received from the printing apparatus **101**. FIG. **17**B illustrates attribute information about the print job transmitted from the CPS **102** to the printing apparatus **101**. The CPS **102** does not discard attribute information that is not supported by the CPS **102**, and directly transfers the attribute information to the printing apparatus **101**. Accordingly, the printing apparatus **101** can appropriately interpret the attribute information received via the CPS **102** and executes printing.
[0125] Next, an alternative sequence to be executed when the corresponding print extension application is not provided will be described. In step S**1530**, the client terminal extracts attribute information supported by the print client from the attribute information acquired in step S**1511**. Then, the output destination printer information in which the extracted attribute information is associated with the output destination is registered, and then the series of processing is terminated.
[0126] Specific control processing in the client terminal will be described with reference to FIG. **18**. Each operation (step) in the flowchart illustrated in FIG. **18** is implemented by the CPU **131** of the client terminal **104** when the CPU **131** executes control programs stored in the ROM **132** or the storage **134**. If there is a need to clarify a main program that executes the processing, a control program (OS or extension application) to be executed by a processor is described as the subject.
[0127] The flowchart illustrated in FIG. **18** is executed when an operation for selecting a printer that is not registered in the client terminal **104** but is registered on the CPS **102** from among the printers found by printer search processing is received.
[0128] In step S**1801**, the OS of the client terminal **104** sends an inquiry about whether the print extension application corresponding to the output destination printer to be registered is present to the distribution server. As a result of the inquiry, if the corresponding print extension application is present (YES in step S**1801**), the processing proceeds to step S**1802**. If the print extension application is not present (NO in step S**1801**), the processing proceeds to step S**1104**. In step S**1802**, the OS of the client terminal **104** downloads the print extension application from the distribution server and installs the print extension application in the client terminal **104**. After the installation is completed, the processing proceeds to step S**1803**.
[0129] In step S**1803**, the OS of the client terminal **104** registers, on the print framework, output destination printer information in which the attribute information including attribute information that is supported by the client and attribute information that is not supported by the client is associated with the output destination. Next, in step S**1804**, the OS of the client terminal **104** stores the correspondence relation between the print extension application and the output destination printer in the registry DB. For example, an application path for starting the print extension application and information for identifying the output destination printer are stored in association with each other.
[0130] In step S**1805**, the OS of the client terminal **104** sets the printer for which the registration processing is completed as the output destination, and displays the print setting screen generated based on the attribute information supported by the print client. In this case, the print settings corresponding to the attribute information that is not supported by the print client are not displayed.
[0131] In step S**1806**, the OS of the client terminal **104** determines whether a user instruction for making advanced print settings has been made on the print setting screen displayed in step S**1805**. If it is detected that the user instruction for making advanced print settings has been made (YES in step S**1806**), the processing proceeds to step S**1807**. If it is not detected that the user instruction for making advanced print settings has not been made (NO in step S**1806**), the processing proceeds to

step S**1811**.

[0132] In step S**1807**, the OS of the client terminal **104** determines whether the print extension application corresponding to the output destination printer is present. Specifically, the OS acquires information for identifying the output destination printer, and determines whether the information for identifying the print extension application corresponding to the output destination printer is stored in the registry DB. If the information for identifying the print extension application corresponding to the output destination printer is stored (YES in step S**1807**), the processing proceeds to step S**1808**. On the other hand, if the information for identifying the print extension application corresponding to the output destination printer is not stored (NO in step S**1807**), the advanced settings screen is not displayed and the processing proceeds to step S**1811**.

[0133] In step S**1808**, the OS of the client terminal **104** starts the print extension application. The print extension application started by the OS acquires attribute information included in the output destination printer information, and displays the advanced print settings screen illustrated in FIG. **16** based on the attribute information. The user can change the print settings via the setting screen. The print extension application that has received a setting change operation updates the display of the setting screen and feeds the current setting state back to the user.

[0134] In step S**1809**, the print extension application determines whether a user operation for completing the print setting has been received. If the user operation for completing the print setting has been received (YES in step S**1809**), the processing proceeds to step S**1810**. If the user operation for completing the print setting has not been received (NO in step S**1809**), the print extension application waits for a print setting operation to be made via the screen of the print extension application.

[0135] In step S**1810**, the print extension application updates the attribute information managed by the OS based on the print setting that is made via the screen and is to be applied to the print job. After completion of the updating, the print extension application interrupts the operation of the print extension application itself. The print client of the OS that has detected that the operation of the print extension application is interrupted displays the print setting screen based on the updated print setting. At this time, the print client ignores attribute information that is not supported by the print client, and displays the print setting screen.

[0136] In step S**1811**, the OS determines whether a print start instruction has been made via the print setting screen. If the print start instruction has been made (YES in step S**1811**), the processing proceeds to step S**1812**. If the print start instruction has not been made (NO in step S**1811**), the processing returns to step S**1806** to wait for a print setting operation.

[0137] In step S**1812**, the print client of the client terminal **104** generates a print job and transmits the generated print job to the CPS **102**. It is assumed that the print job is generated by Create-Job operation or Print-Job operation defined in the IPP. In a case where the instruction is made by Print-Job operation, both of the attribute information and print data are transmitted. If the instruction is made by Create-Job operation, the print job is generated on the CPS **102** by Create-Job operation including print attributes. Then, processing for associating the generated print job with at least one piece of document data is performed using an operation for adding a document to be printed. At this time, the print client also transmits attribute information that is not supported by the print client to the CPS **102** as attribute information related to the print job. Accordingly, the attribute information updated by the print extension application is not discarded by the print client but is transmitted to the CPS **102**. The processing described above makes it possible to support more flexible print settings.

MODIFIED EXAMPLES

[0138] The present exemplary embodiment illustrates an example where print settings are made during print processing via the CPS. However, the present exemplary embodiment is not limited to this example. For example, the present exemplary embodiment can also be applied to a scan setting when scan data obtained by an MFP is transmitted to a personal computer (PC) of the user via the

cloud. In this case, the MFP notifies a cloud scanning service of scanning capabilities corresponding to the full specification of the MFP when the MFP is registered on the cloud scanning service. At this time, the MFP notifies the cloud scanning service of capabilities including attributes unique to the vendor itself. The cloud scanning service registers a scanning capability that is not supported by the cloud scanning service itself on the cloud. A scanning client installed in the client terminal that uses the cloud scanning service acquires capability information corresponding to the full specification from the cloud scanning service. Then, the scanning client extracts the capability information supported by the scanning client from the full specification capability information and registers the extracted scanning capability information on a scanning framework. Assume that, when cloud scanning is applied to the second exemplary embodiment, a scanning extension application is downloaded from the distribution server and the downloaded scanning extension application is installed. In this case, high-performance optical character recognition (OCR) processing provided by the MFP, a scanned content summarization service provided by an artificial intelligence (AI) engine of the MFP, and the like can be set from an extension print setting screen for the scanning extension application.
OTHER EMBODIMENTS
[0139] Embodiments of the present disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions recorded on a storage medium (e.g., non-transitory computer-readable storage medium) to perform the functions of one or more of the above-described embodiment(s) of the present disclosure, and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more of a central processing unit (CPU), micro processing unit (MPU), or other circuitry, and may include a network of separate computers or separate computer processors. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.
[0140] While the present disclosure includes exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

Claims

1. A client terminal connectable with a print service that communicates with a printer using IPP (Internet Printing Protocol) and that has registered information of the printer including information of a first attribute defined in a specification of the IPP and information of a second attribute not defined in the specification of the IPP, the client terminal comprising: a controller including a processor and a memory, the controller configured to: enable a user to set first print setting information that is based on the information of the first attribute defined in the specification of the IPP and second print setting information that is based on the information of the second attribute not defined in the specification of the IPP; and provide the set first and second print setting information to be used for printing.
2. The client terminal according to claim 1, wherein, the controller is further configured to receive the information of the first attribute and the information of the second attribute from the print service.
3. The client terminal according to claim 1, wherein the set first and second print setting

information is provided to the print service.

4. A method for controlling a client terminal connectable with a print service that communicates with a printer using IPP (Internet Printing Protocol) and that has registered information of the printer including information of a first attribute defined in a specification of the IPP and information of a second attribute not defined in the specification of the IPP, the method comprising: enabling a user to set first print setting information that is based on the information of the first attribute defined in the specification of the IPP and second print setting information that is based on the information of the second attribute not defined in the specification of the IPP; and providing the set first and second print setting information to be used for printing.

5. The method according to claim 4, further comprising: receiving the information of the first attribute and the information of the second attribute from the print service.

6. The method according to claim 4, wherein the set first and second print setting information is provided to the print service.

7. A non-transitory computer-readable medium storing executable instructions, which when executed by one or more processors of a client terminal connectable with a print service that communicates with a printer using IPP (Internet Printing Protocol) and that has registered information of the printer including information of a first attribute defined in a specification of the IPP and information of a second attribute not defined in the specification of the IPP, cause the client terminal to perform operations comprising: enabling a user to set first print setting information that is based on the information of the first attribute defined in the specification of the IPP and second print setting information that is based on the information of the second attribute not defined in the specification of the IPP; and providing the set first and second print setting information to be used for printing.

8. The non-transitory computer-readable medium according to claim 7, the operations further comprising: receiving the information of the first attribute and the information of the second attribute from the print service.

9. The non-transitory computer-readable medium according to claim 7, wherein the set first and second print setting information is provided to the print service.